A journey with Anthropic researchers

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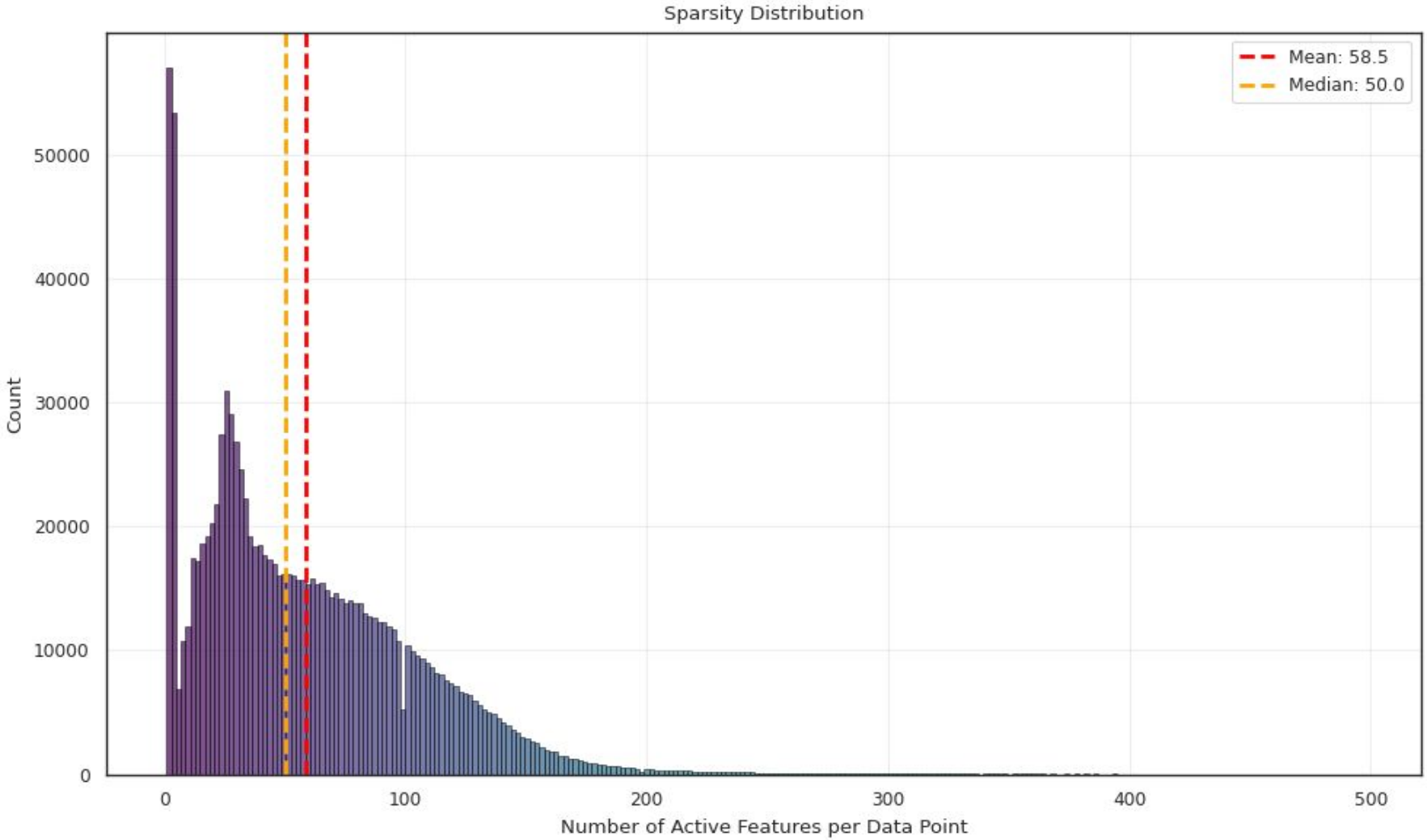
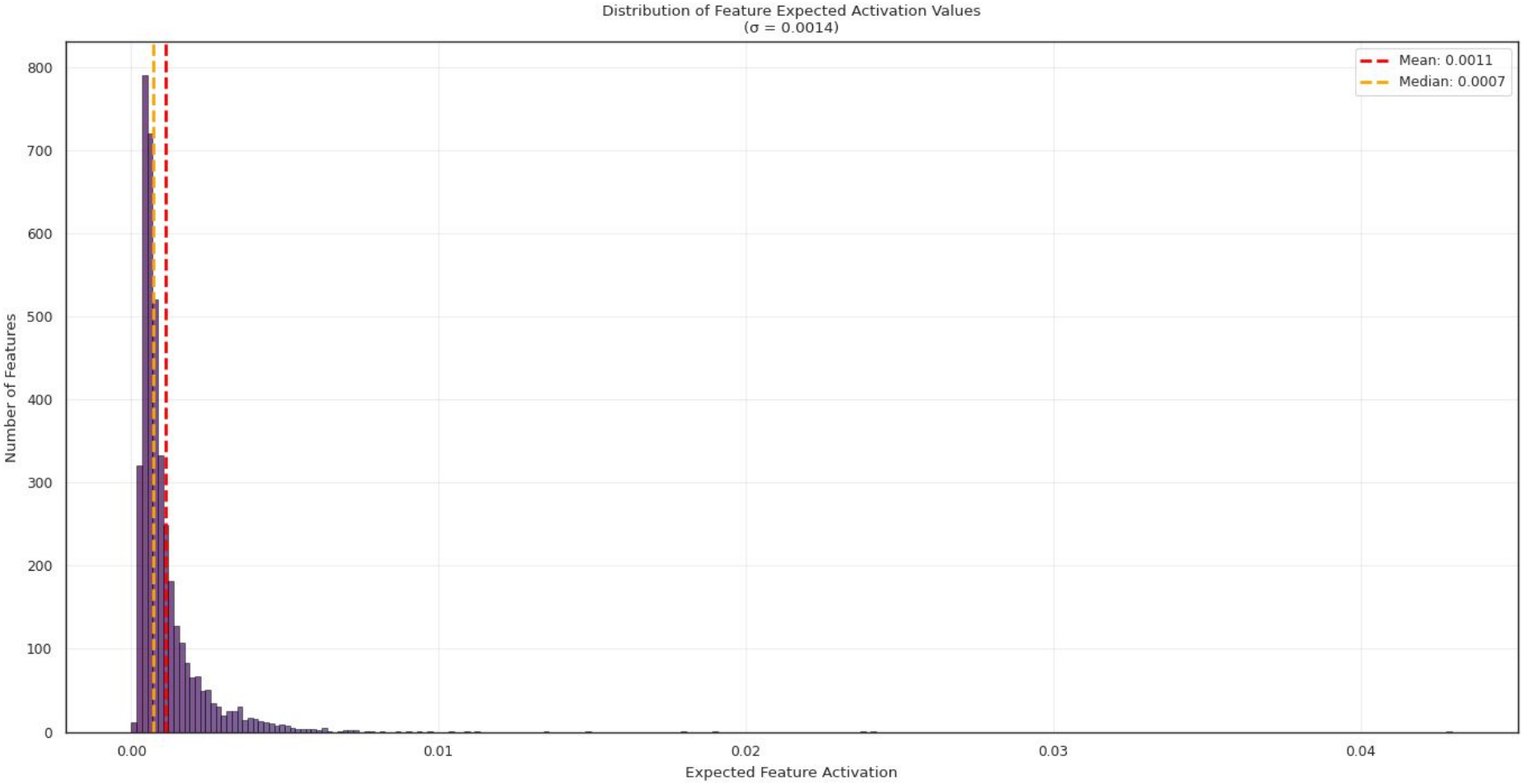
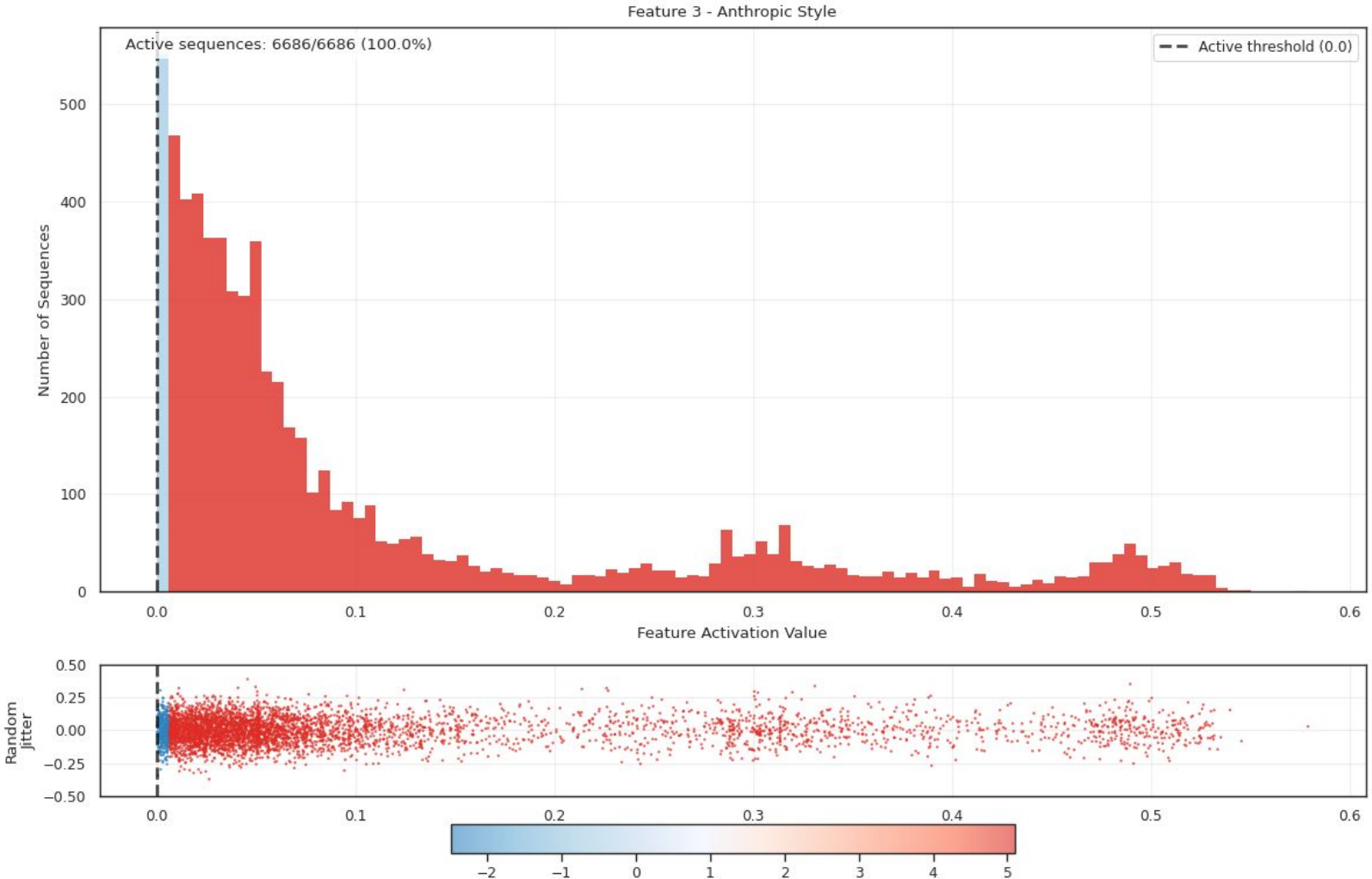
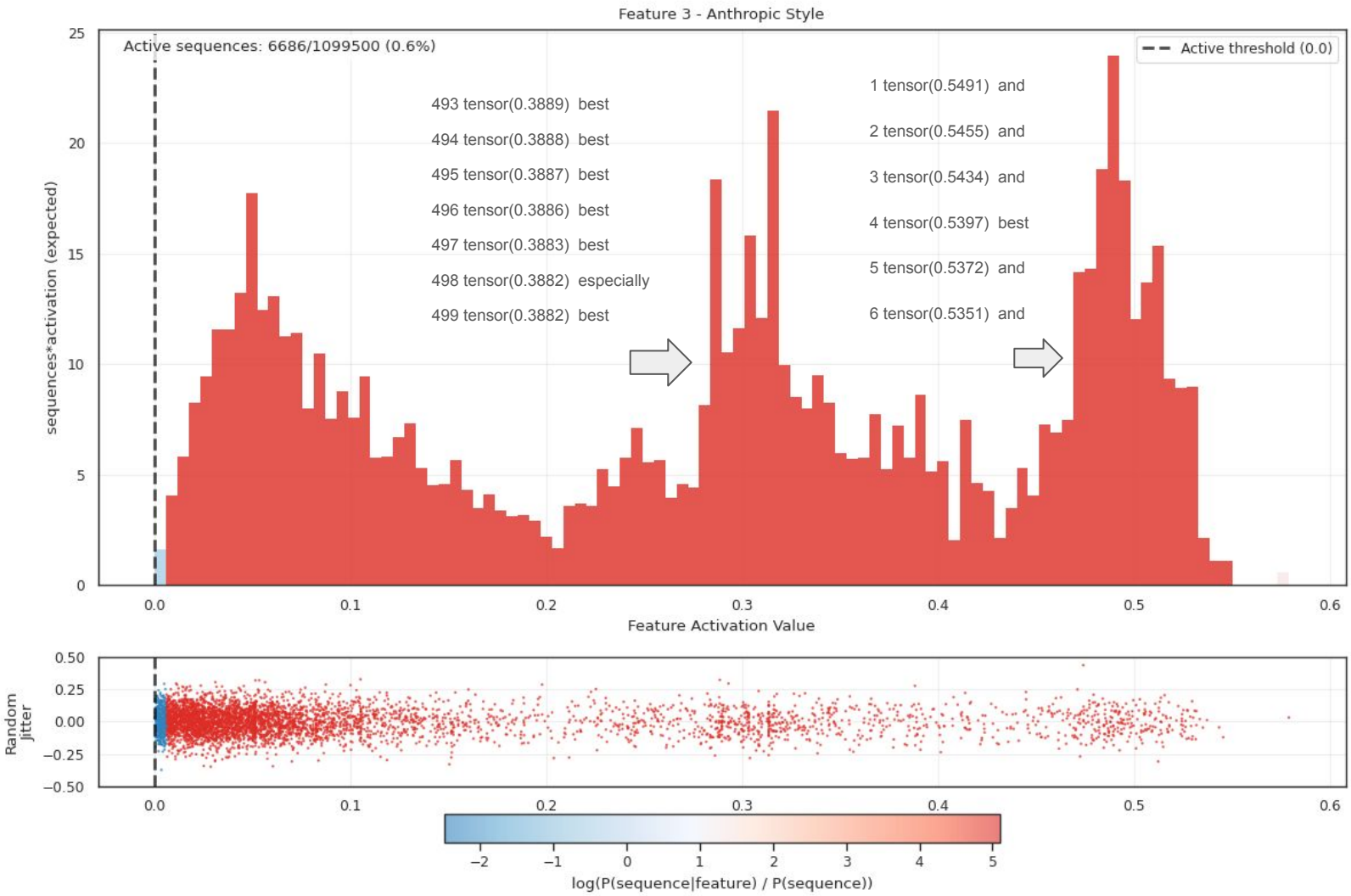
What We've Achieved

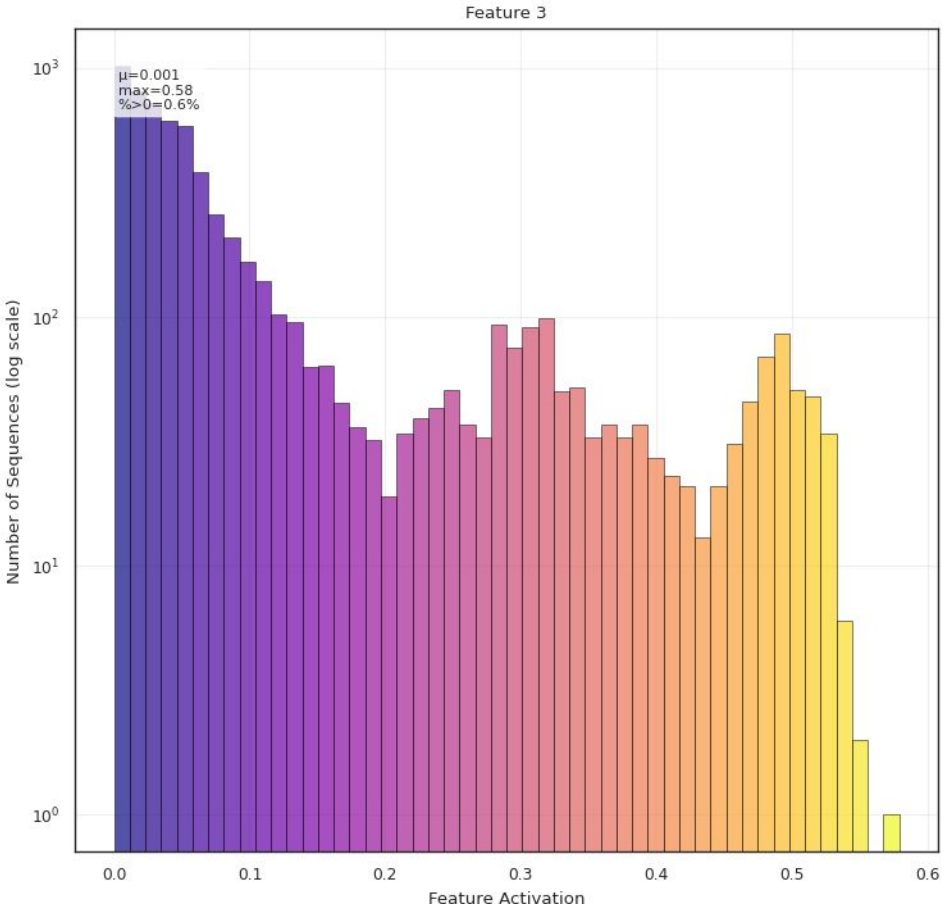
- train sparse transcoder
- feature visualization system in “Anthropic style”

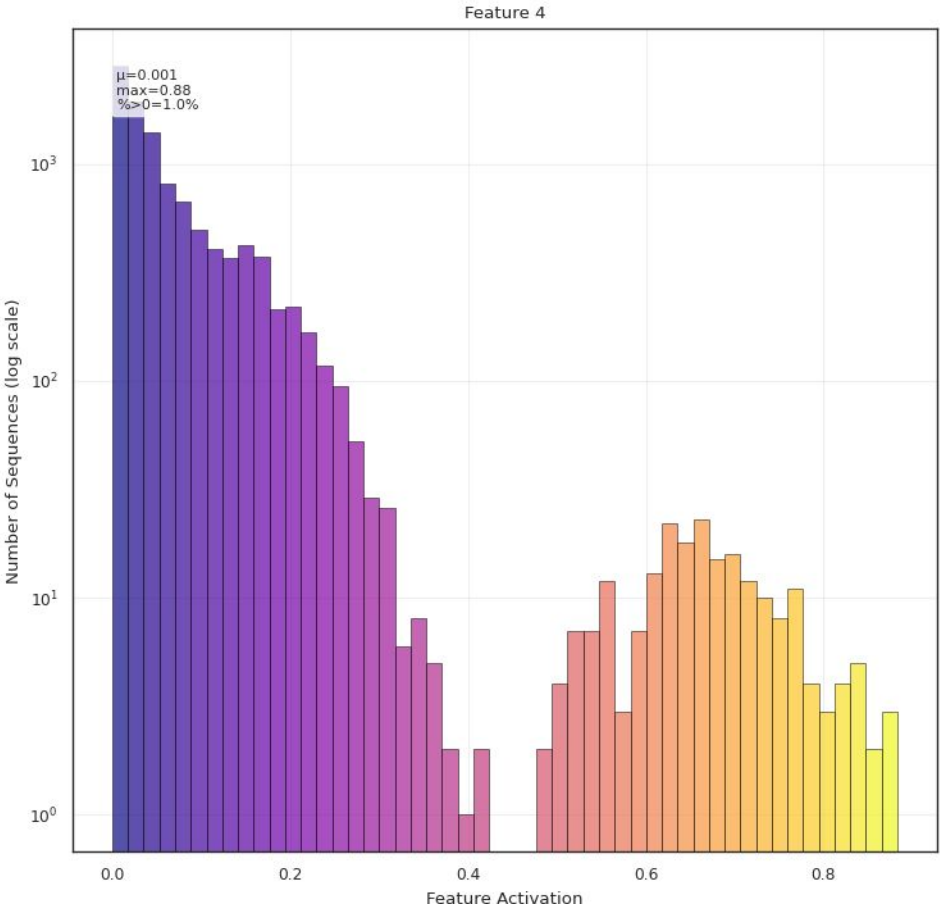

Sparsity Distribution
Mean: 58.5
Median: 50.0
Count
0
10000
20000
30000
40000
50000
0
100
200
300
400
500
Number of Active Features per Data Point

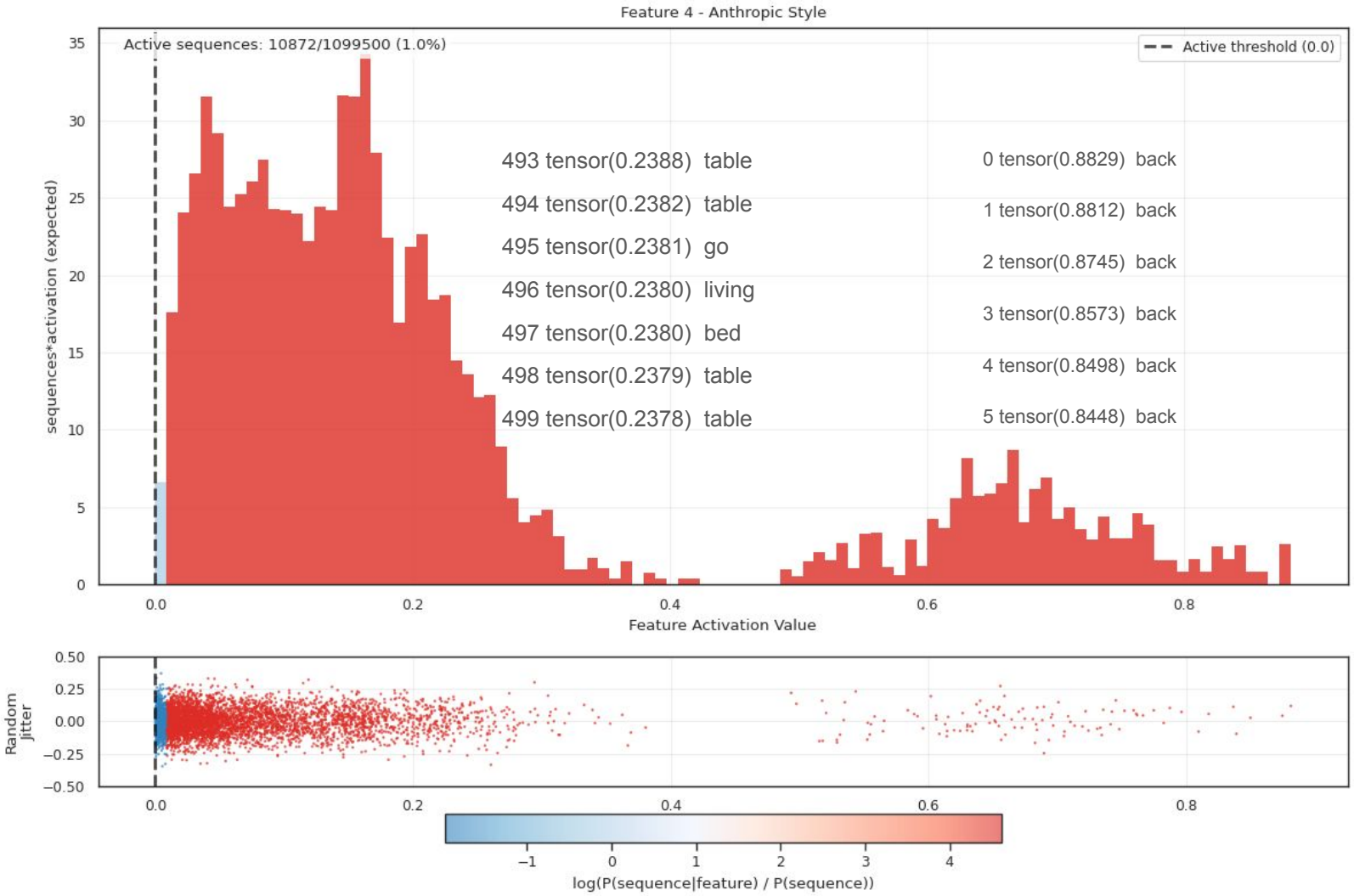

Distribution of Feature Expected Activation Values
(σ = 0.0014)
Mean: 0.0011
Median: 0.0007
Number of Features
Expected Feature Activation
0
100
200
300
400
500
600
700
800
0.00
0.01
0.02
0.03
0.04

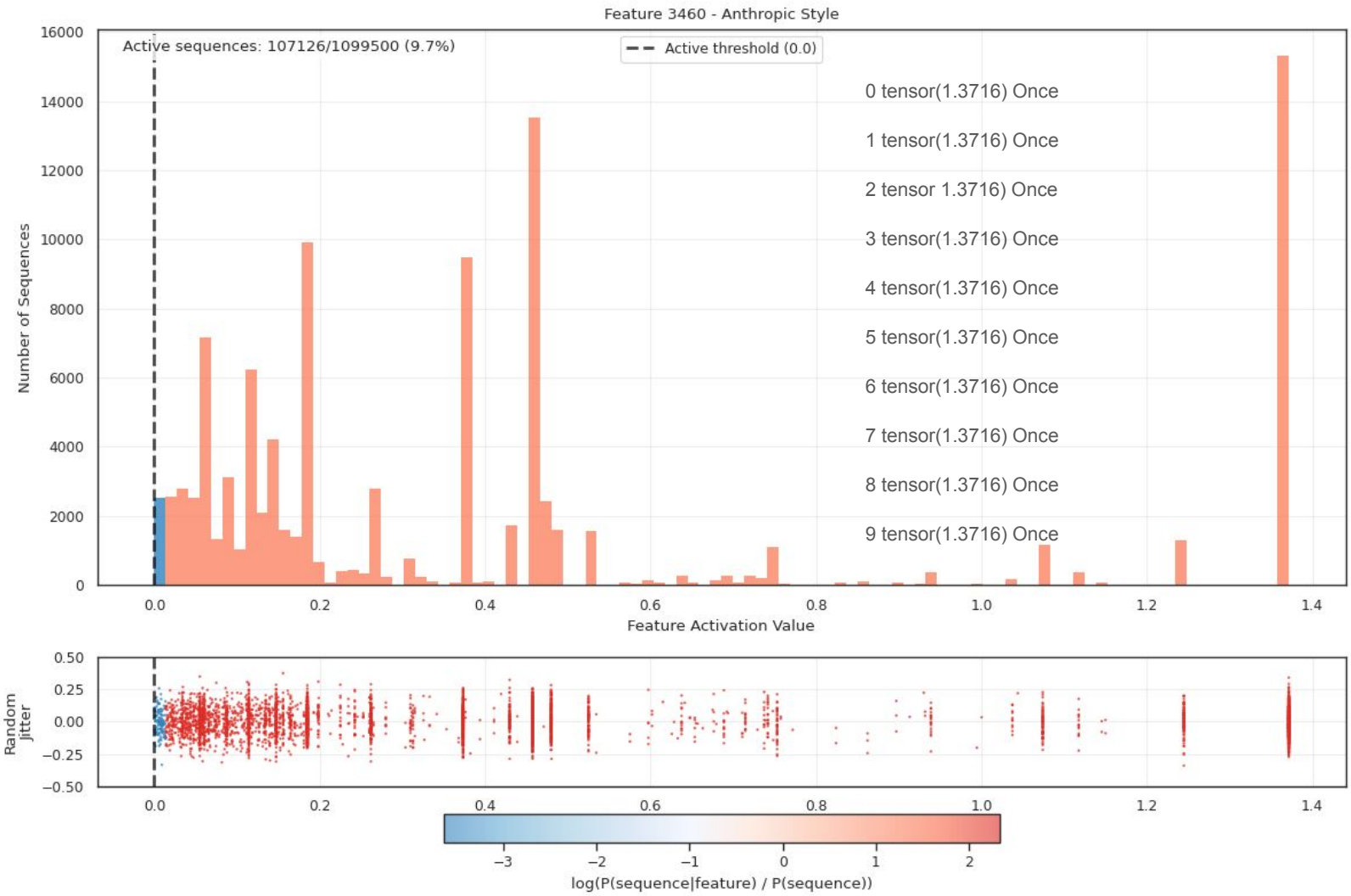

Feature 3 - Anthropic Style
Active sequences: 6686/6686 (100.0%)
Active threshold (0.0)
Number of Sequences
Feature Activation Value
Random Jitter

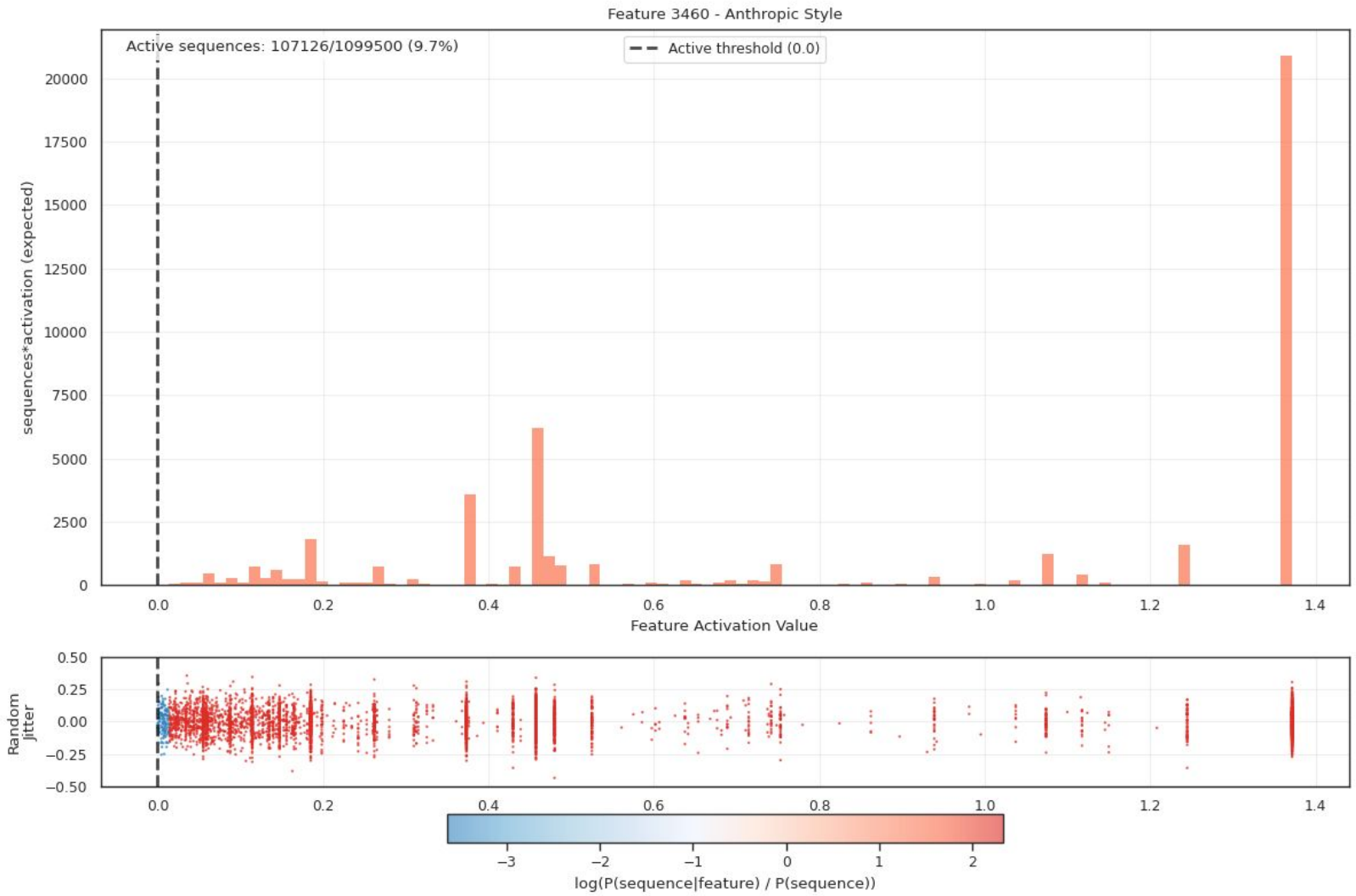

Feature 3 - Anthropic Style
Active sequences: 6686/1099500 (0.6%)
Active threshold (0.0)
493 tensor(0.3889) best
494 tensor(0.3888) best
495 tensor(0.3887) best
496 tensor(0.3886) best
497 tensor(0.3883) best
498 tensor(0.3882) especially
499 tensor(0.3882) best
1 tensor(0.5491) and
2 tensor(0.5455) and
3 tensor(0.5434) and
4 tensor(0.5397) best
5 tensor(0.5372) and
6 tensor(0.5351) and
sequences*activation (expected)
Feature Activation Value
Random jitter
log(P(sequence|feature) / P(sequence))

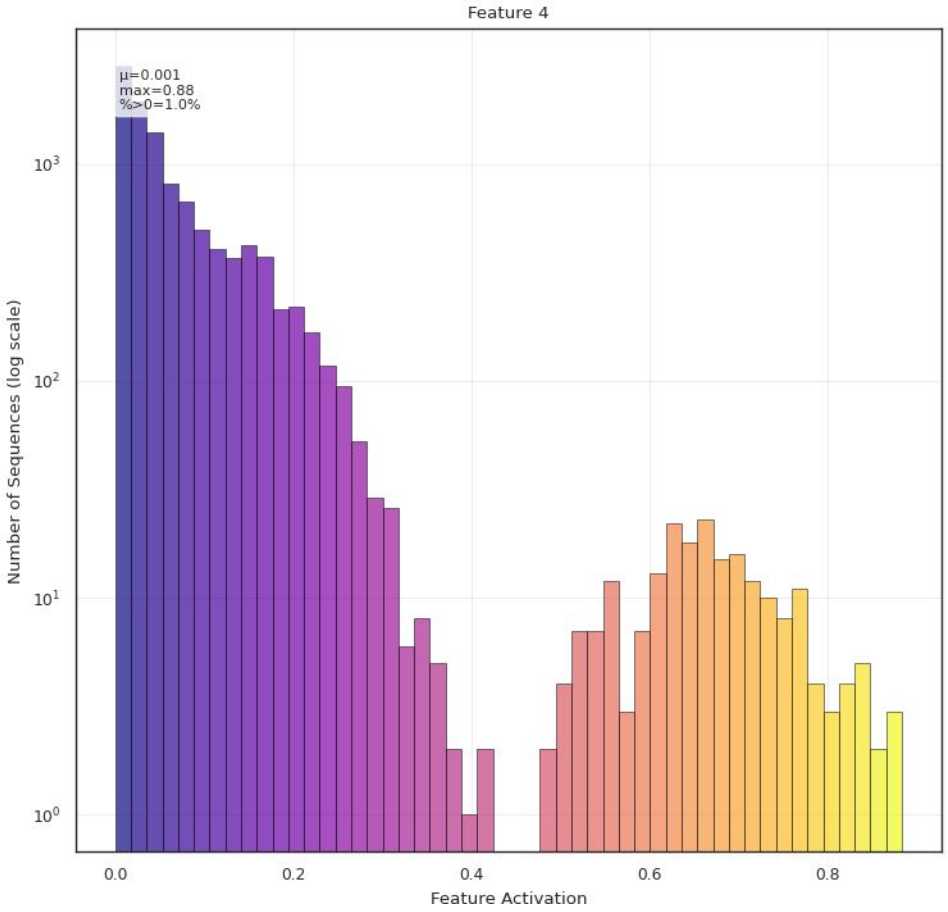

Feature 3
μ=0.001
max=0.58
%>0=0.6%
Number of Sequences (log scale)
Feature Activation
0.0
0.1
0.2
0.3
0.4
0.5
0.6

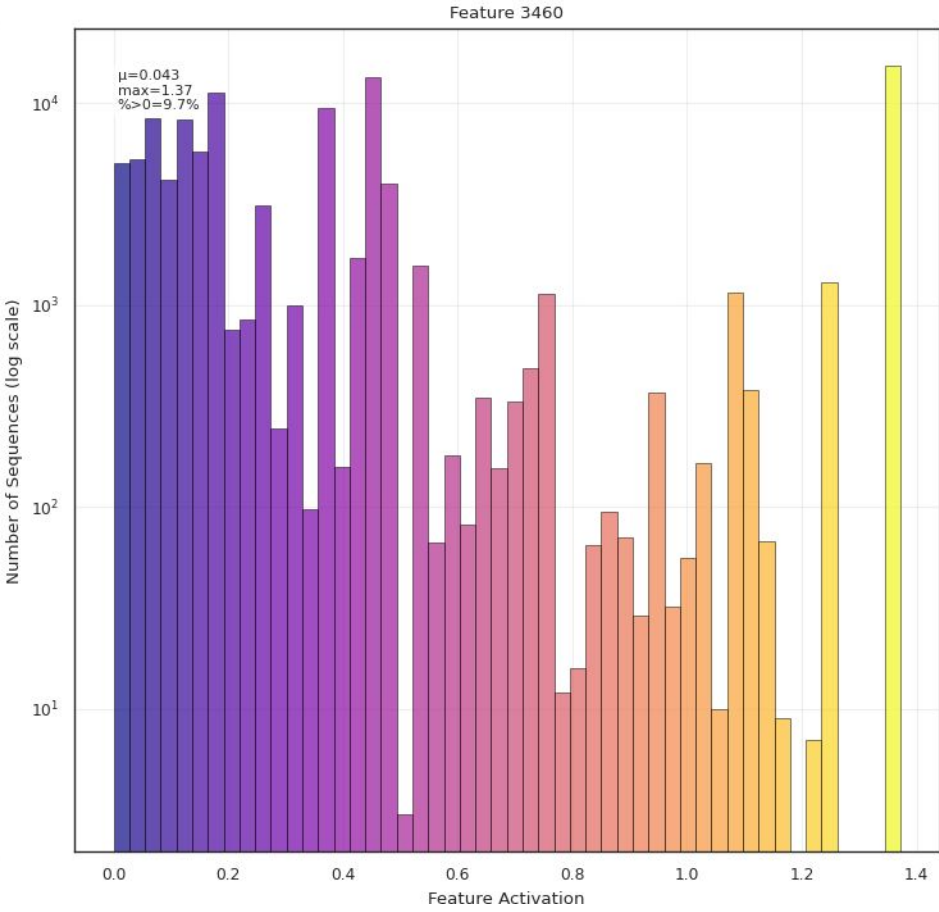

Feature 4
μ=0.001
max=0.88
%>0=1.0%
Number of Sequences (log scale)
Feature Activation
0.0
0.2
0.4
0.6
0.8

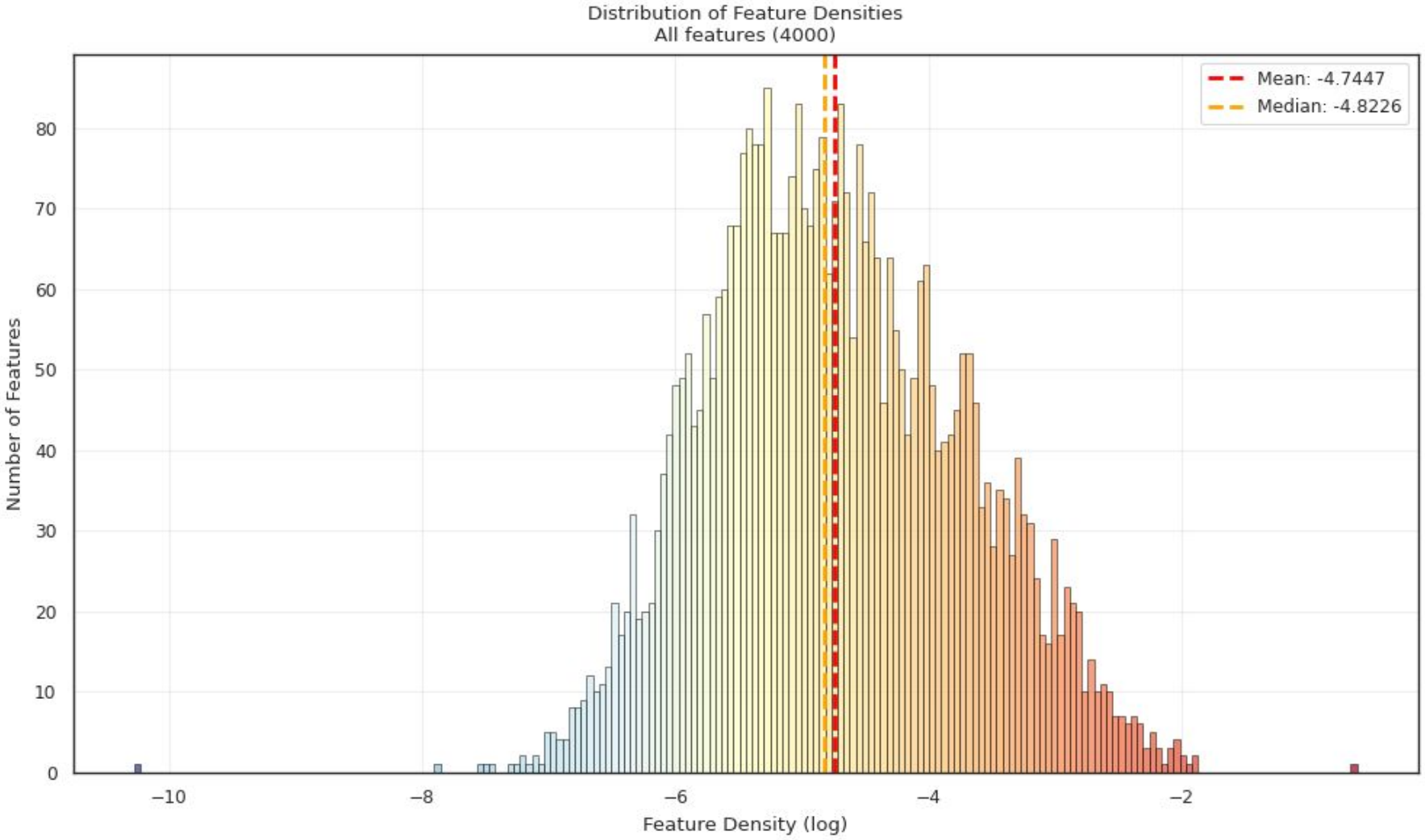

Feature 4 - Anthropic Style
Active sequences: 10872/1099500 (1.0%)
Active threshold (0.0)
493 tensor(0.2388) table
494 tensor(0.2382) table
495 tensor(0.2381) go
496 tensor(0.2380) living
497 tensor(0.2380) bed
498 tensor(0.2379) table
499 tensor(0.2378) table
0 tensor(0.8829) back
1 tensor(0.8812) back
2 tensor(0.8745) back
3 tensor(0.8573) back
4 tensor(0.8498) back
5 tensor(0.8448) back
sequences*activation (expected)
Feature Activation Value
Random jitter
log(P(sequence|feature) / P(sequence))

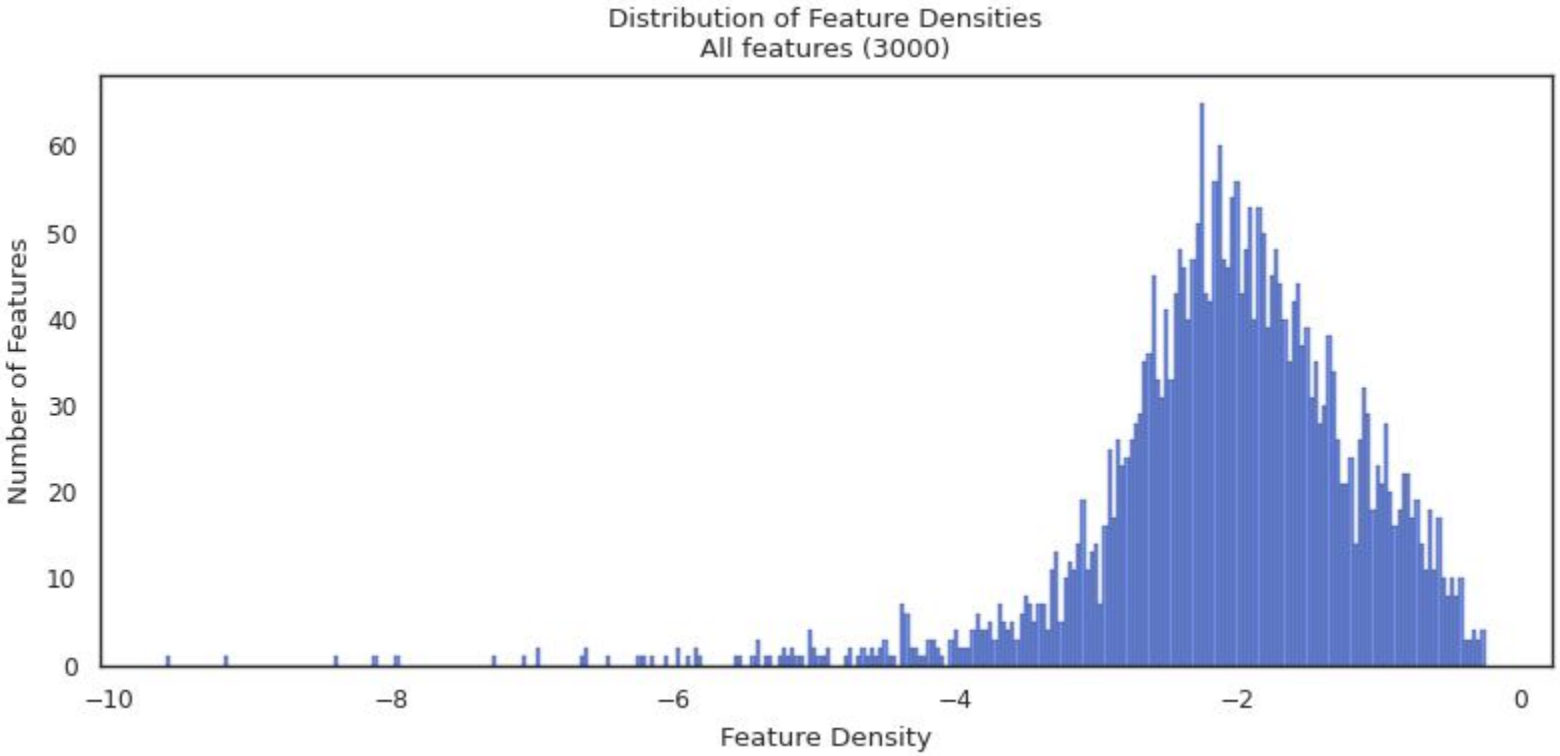

Feature 3460 - Anthropic Style
Active sequences: 107126/1099500 (9.7%)
Active threshold (0.0)
0 tensor(1.3716) Once
1 tensor(1.3716) Once
2 tensor 1.3716) Once
3 tensor(1.3716) Once
4 tensor(1.3716) Once
5 tensor(1.3716) Once
6 tensor(1.3716) Once
7 tensor(1.3716) Once
8 tensor(1.3716) Once
9 tensor(1.3716) Once
Number of Sequences
Feature Activation Value
Random jitter
log(P(sequence|feature) / P(sequence))


Feature 3460 - Anthropic Style
Active sequences: 107126/1099500 (9.7%)
Active threshold (0.0)
sequences*activation (expected)
Feature Activation Value
Random jitter
log(P(sequence|feature) / P(sequence))


Feature 4
μ=0.001
max=0.88
%>0=1.0%
Number of Sequences (log scale)
Feature Activation


Feature 3460
μ=0.043
max=1.37
%>0=9.7%
Number of Sequences (log scale)
Feature Activation


Distribution of Feature Densities
All features (4000)
Mean: -4.7447
Median: -4.8226
Number of Features
Feature Density (log)
0
10
20
30
40
50
60
70
80
−10
−8
−6
−4
−2


Distribution of Feature Densities
All features (3000)
Number of Features
0
10
20
30
40
50
60
−10
−8
−6
−4
−2
0
Feature Density

Visualization demo (case study)

- feature analysis created by LLM
- notebooks

What’s next?

- train a full crosscoder
- code for generating pruned attribution graphs

Thank you!